

Internship Project Assignment – Mugisha Yves

Project Title: Company Internal Reporting System

Project Objective

The objective of this project is to design and develop a full-stack **Internal Reporting System** for the company. The system will allow employees to submit periodic reports based on schedules defined by the admin, with a structured approval workflow and full tracking of submissions.

The project must follow structured software development practices, starting with system documentation, then implementation using **Next.js (frontend)** and **Node.js (backend)**. All work must be managed using GitHub.

Phase 1: System Design and Documentation

Before development begins, you must prepare and submit:

1. Software Requirements Specification (SRS)

The SRS must include:

- System overview
- User roles and permissions
- Functional requirements
- Non functional requirements
- System modules
- Use cases

2. Data Flow Diagrams (DFDs)

Provide:

- Context diagram
- Level 1 DFD
- Level 2 DFD where necessary

3. Entity Relationship Diagram (ERD)

Include:

- Entities such as Users, Departments, Teams, Reports, Reviews
- Relationships between entities
- Primary and foreign keys

Approval of these documents is required before proceeding to development.

Phase 2: System Development

Technologies to Use

Frontend
Next.js (React framework)

Backend
Node.js with Express

Database Options
PostgreSQL
MySQL

System Overview

The system will manage:

Employees assigned to departments and teams
Report submissions based on defined schedules
A two-stage approval workflow

Core Features

1. User and Organization Management

Admin should:

Create departments
Create teams within departments
Register employees
Assign employees to departments and teams

Roles include:

Admin
Employee
Department Reviewer
Final Approver (e.g., COO)

2. Report Scheduling

Admin should:

Define report frequency (monthly, bi-weekly, etc.)
Set report types per department/team
Automatically generate reporting timelines

3. Report Submission

Employees should:

- Submit reports based on assigned schedule
- Upload files or fill report forms
- Track submission status

4. Review and Approval Workflow

Stage 1:

Each department has one assigned reviewer

Stage 2:

Reports are sent to a central approver (e.g., COO)

Actions available:

- Approve
- Reject
- Request changes

5. Notification System

System must notify users when:

- Report is requested
- Report is submitted
- Report is reviewed
- Report is approved or rejected

Notifications include:

- In-app alerts
- Email notifications

6. Report Tracking and Records

System should:

- Store all reports with timestamps
- Maintain status history
- Allow filtering by department, date, and status

Statuses include:

- Pending
- Under Review

Approved
Rejected

7. Dashboard and Analytics

Admin dashboard:

Overview of all reports
Compliance tracking

Department dashboard:

Submission progress

Reviewer dashboard:

Pending approvals

8. Search and Filtering

Allow filtering by:

Department
Team
Date
Report type
Status

Backend Requirements

Develop RESTful APIs for:

User authentication
Department and team management
Report submission
Review and approval workflow
Notifications

Include validation and proper error handling.

Frontend Requirements

Responsive UI using Next.js
Role-based dashboards
Forms for report submission
Tables for tracking reports
Status indicators

GitHub Version Control Requirements

Create a GitHub repository

Use branches:

main
dev

Commit regularly with clear messages
Push work continuously

Maintain a README file with:

Project description
Setup instructions
Technologies used

Expected Deliverables

Approved SRS document
DFDs and ERD
Fully working system
Frontend and backend implementation
Database integration
Notification system
GitHub repository
Documentation

Acceptance Criteria

Documentation is complete and approved
Users can log in and perform role-based actions
Reports follow full workflow
Approval process works correctly
Notifications are triggered properly
System runs without major errors
Code is clean and structured
GitHub shows continuous development

Project Timeline

Minimum: 1 week
Maximum: 2 weeks

All progress must be pushed continuously to GitHub during development.